MILITARY SPECIFICATION SHEET

CONTACT, ELECTRICAL CONNECTOR,
CONCENTRIC TWINAX, PIN, SHIELDED, SIZE 8
(FOR DOD-C-83527 CONNECTOR)

This military specification has been prepared by the Electronic Support Division, AFLC 2750 ABW/ES based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-39029/96. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

The complete requirements for acquiring the contact described herein shall consist of this specification and the latest issue of MIL-C-39029.

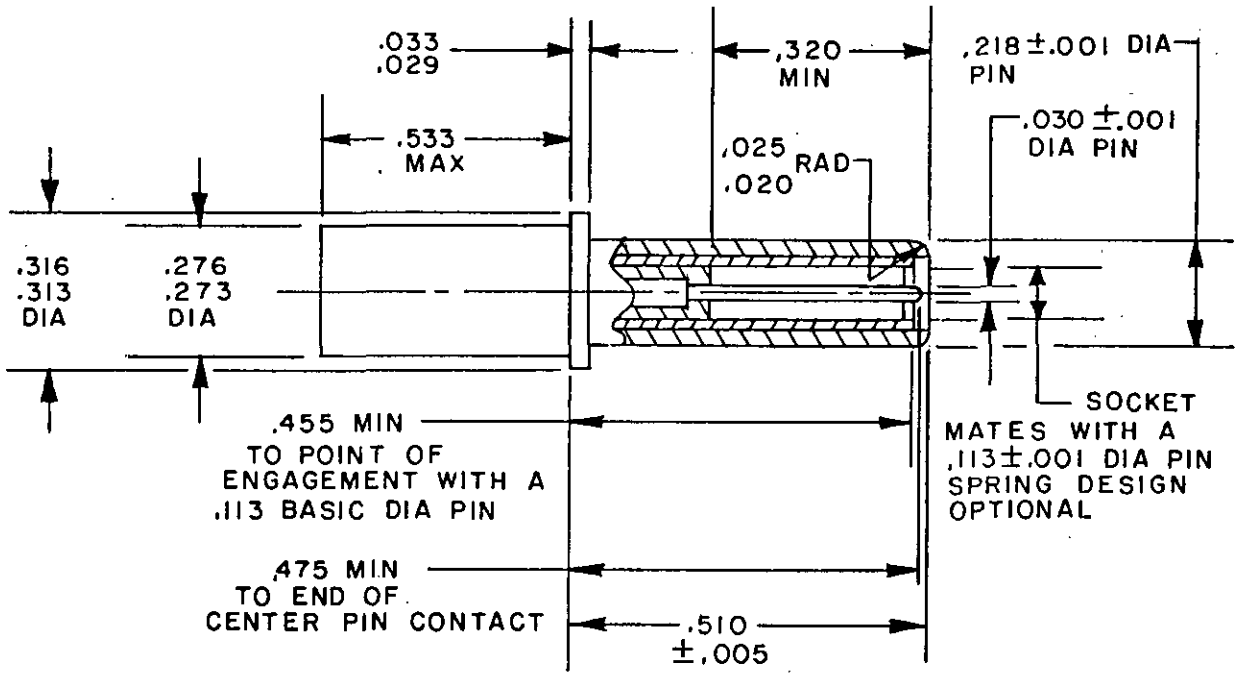
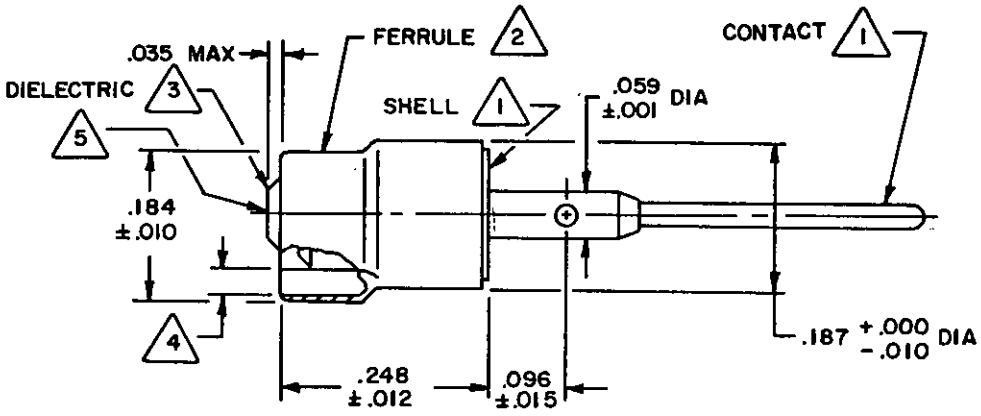


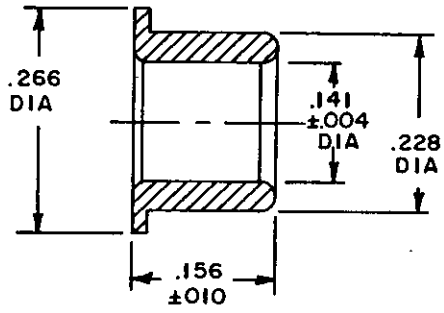
FIGURE 1. Concentric twinax contact.

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1. Brass per QQ-B-626.
2. Copper per QQ-C-576.
3. PTFE per MIL-P-19468.
4. .0285 gage pin must enter for a length of .085 min.
5. .028 GO - .031 NO GO. Gage will enter to depth of sight hole.

CRIMP END ASSEMBLY, PIN (INNER)

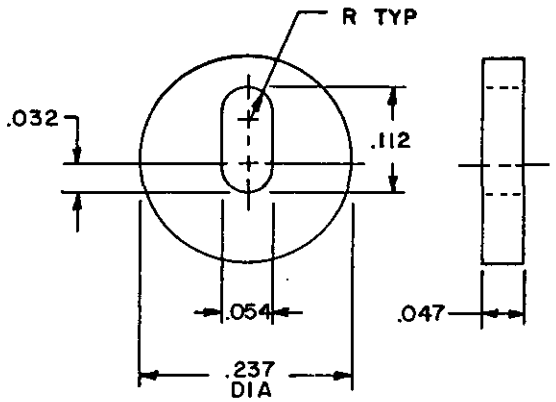


NOTE: Brass per QQ-B-626.

INNER FERRULE

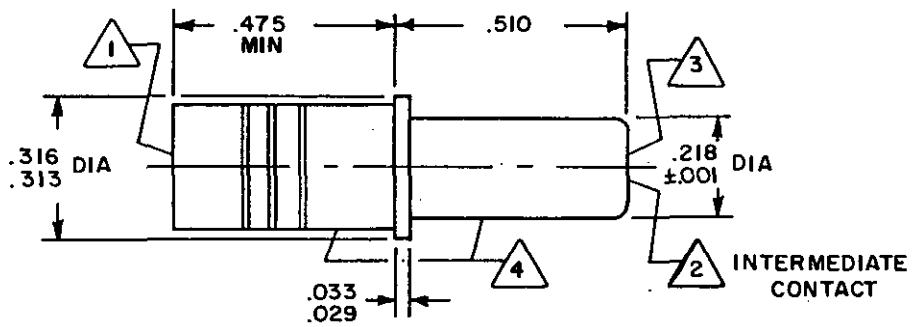
FIGURE 1. Concentric twinax contact - continued.

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NOTE: PTFE per MIL-P-19468.

BUSHING, INSULATING.



1 Press ring (internal), brass per QQ-B-626.
2 Beryllium copper per QQ-C-530.
3 PTFE per MIL-P-19468.
4 Brass per QQ-B-626.

PIN ASSEMBLY

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.0285	.724	.054	1.37	.184	4.67	.276	7.01
.004	.10	.029	.74	.059	1.50	.187	4.75	.313	7.95
.005	.13	.030	.76	.085	2.16	.218	5.54	.316	8.03
.010	.25	.031	.79	.096	2.44	.228	5.79	.320	8.13
.012	.30	.032	.81	.112	2.84	.237	6.02	.455	11.56
.015	.38	.033	.84	.113	2.87	.248	6.30	.475	12.07
.020	.51	.035	.89	.141	3.58	.266	6.76	.485	12.32
.025	.64	.047	1.19	.156	3.96	.273	6.93	.510	12.95
.028	.71							.533	13.54

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. Tolerance for .xxx decimals ±.005 unless otherwise specified.

FIGURE 1. Concentric twinax contact - continued.

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ENGINEERING DATA:

Frequency range: 0 to 20 MHz.

Voltage rating: 500 volts rms maximum; working voltage at sea level, 125 volts rms maximum at 70,000 feet.

Temperature rating: -65°C to +150°C.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Tools: See tables II and III.

Assembly procedure: Assembly instructions to be developed. Manufacturer's recommended assembly instructions shall be shipped with unit package.

Mating contact: See MIL-C-39029/95.

Low signal level contact resistance (center contact only): See table IV.

Contact resistance: See table V.

Contact engagement and separation force: The engagement depth shall be as encountered in normal service. The test pins shall be in accordance with MS3197, except the diameters shall be as specified in table VI.

Crimp tensile strength (center, intermediate, and outer contact crimp joint): See table VII.

High impact shock: All contacts shall be wired in a series circuit with 100 milliamperes maximum current flow through the series circuit during shock. Connectors shall be monitored for any discontinuities. A detector capable of detecting all discontinuities in excess of 1 microsecond shall be used. Wired and mated connectors shall be subjected to the test specified in MIL-S-901, grade A with the following modifications and additions. Mounting fixture shall be in accordance with MIL-S-901, lightweight. The cable or wire bundle shall be supported on a stationary frame to provide a free flexing cable length between the frame and fixture of not less than 36 inches (914.4 mm).

a. Coupling: Coupled by normal mounting means.

b. Test conditions A: Plug shall be terminated with at least 80 percent of wired contacts. The wire bundle shall be provided with straight, open frame, strain relief accessory hardware.

Insulation resistance: Applicable.

Corrosion (salt spray): Applicable.

Humidity: Method 1002, type II, per MIL-STD-1344.

Dielectric withstanding voltage: See table VIII.

Durability: Applicable.

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Thermal shock: High temperature shall be +85°C except for connectors using +200°C cables; high temperature shall then be +150°C.

Permeability: Applicable.

Temperature life: Applicable.

Plating porosity: Applicable.

Part number: M39029/96-546.

QPL evaluating activity: Defense Electronics Supply Center (DESC-E), Dayton, OH 45444.

Contacts shall comply with reliability assurance provisions of MIL-STD-790 as specified in MIL-C-38999.

TABLE I. Design characteristics

BIN code	Color bands			Contact cavity size	Cable accommodated	Type	Class
	1st	2nd	3rd				
546	Green	Yellow	Blue	8	M17/176-00002	E	B

TABLE II. Tools (installing and removal).

BIN code	Installing and removal tool
546	M81969/28-03

TABLE III. Tools (crimping).

BIN code	Basic crimping tool	Die part no.
546	M22520/5-01	M22520/5-104

TABLE IV. Low signal level contact resistance (center contact only).

BIN code	Cable accommodated	Maximum contact resistance (milliohms)	
		Initial	After conditioning
546	M17/176-00002	55	66

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TABLE V. Contact resistance.

BIN code	Contact	Cable accomodated	Test current (amperes)	Maximum voltage drop (millivolts)		
				25°C, +3°C, -0°C		+200°C, +3°C, -0°C
				Initial	After conditioning	After conditioning
546	Center	M17/176-00002	1.0	15	30	45
	Intermediate		1.0	15	30	45
	Outer		12.0	35	45	70

TABLE VI. Contact engagement and separation force.

Test pin diameter (inch)	Minimum separation force (ounces)		Maximum engagement force (ounces)		Maximum average engagement force
	Initial	After conditioning	Initial	After conditioning	
.1140 +.0000 -.0002	NA	NA	20	24	NA
.1120 +.0002 -.0000	3.0	2.5	NA	NA	NA

TABLE VII. Crimp tensile strength

BIN code	Cable accommodates	Axial load (pounds, minimum)		
546	M17/176-00002	25 (outer braid)	6 (center conducter)	6 (intermediate conductor)

TABLE VIII. Dielectric withstanding voltage.

Contacts	Altitude	Test voltage ac rms
Center to intermediate	Sea level	1000
Intermediate to outer	Sea level	500

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Custodians:
Air Force - 85

Review activities:
Air Force - 11, 15, 17, 99

Agent:
DLA - ES

Preparing activity:
Air Force - 85

(Project 5935-F312-02)

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